

ML Notes1

Machine Learning:

Machine Learning (ML) is the process of training a machine (model) to recognize patterns and make decisions in a way that mimics human thinking — but at a scale and speed no human can match.

Why do we need ML?

When working with massive volumes of data, humans cannot manually identify every pattern or trend. ML models are trained to do this automatically, enabling:

- Pattern recognition in large datasets.
- Predictions on new, unseen data.
- Automation of complex decision-making tasks.

Types of ML Training Approaches:

i) Supervised Learning: Human labels the data; model learns from labeled examples and predicts on new data.

Ex: Spam detection, price prediction

ii) Unsupervised Learning: No labels provided, model finds patterns and clusters on its own.

Ex: Customer segmentation

iii) Semi-Supervised Learning: A portion of data is labeled, model uses labels + unlabeled data to generalize.

Ex: Image classification with few labels

iv) Reinforcement Learning: Model takes actions, gets rewarded for correct predictions, penalized for wrong ones.

Ex: Game playing, robotics

Supervised Learning – Types:

1. Regression Analysis:

Used to predict continuous numerical values.

- Ex: height, weight, house price, salary

2. Classification Analysis:

Used to categorize data into discrete classes.

- Spam or Not Spam

Classification types:

- **Binary Classification:** Only 2 possible outcomes (e.g., Yes/No).
- **Multi-label Classification:** Each item can belong to multiple labels.
- **Multi-class Classification:** More than 2 distinct classes.

ML Workflow:

Data Collection → Data Description → Statistical & Graphical Analysis → Data Transformation → Model Selection (based on EDA) → Standardization & Normalization → Train/Test Split → Model Training → Calculate Model Accuracy → Hyperparameter Tuning → Save Model File → Deployment Strategy → Production Deployment & Testing → Retraining Approach → Logging & Monitoring → Dashboard & Reports

Regression Analysis:

Linear Regression is a supervised learning algorithm that models the relationship between a dependent variable (y) and one or more independent variables (x).

$$Y = mx + c$$

Where: m = slope (coefficient), c = intercept, x = input feature, y = predicted output

Residual Error:

The difference between the actual value and the predicted value is called the residual error. Minimizing this error is the core objective of linear regression.

Residual Error = Actual value – predicted value

Optimization Methods:

1. Ordinary Least Squares (OLS):

- Mathematically finds the best-fit line by minimizing the sum of squared residuals
- No iterative tuning required - produces the global optimal solution directly
- Limitation: No control over the optimization process; can be slow on very large data

2. Gradient Descent:

- An iterative optimization technique used to minimize the loss function
- Starts with random parameters, then takes small steps in the direction that reduces error
- Controlled by the Learning Rate - how big each step is
 - Too large: may overshoot the minimum.
 - Too small: converges slowly or gets stuck in local minima.

Cost Functions (Error Metrics):

i) Mean Absolute Error (MAE): Average of actual – predicted. Simple but sensitive to outliers.

ii) Root Mean Squared Error (RMSE): Square root of average squared errors. Penalizes large errors more heavily.

iii) R-Squared: $1 - (RSS / TSS)$, Measures goodness of fit (0 to 1).

Where:

- **RSS** - Residual Sum of Squares (unexplained variance)
- **TSS** - Total Sum of Squares (total variance in data)
- **ESS** - Explained Sum of Squares ($TSS - RSS$)

An R squared closer to 1 means the model explains the variance well.

Model Fitness – Overfitting vs Underfitting:

i) Well-Trained: Model generalizes correctly. Good accuracy on both train & test.

ii) Overfit: Model memorizes noise in training data. High train accuracy, poor test accuracy. Clean data, regularization, more data.

iii) Underfit: Model is too simple for the complexity. Poor accuracy on both train & test. Use more complex model or features.

Common ML Challenges

Class Imbalance:

Occurs when one class dominates the dataset, leading to biased predictions.

- Ex: 80% spam, 20% not spam - model may always predict 'spam' and still appear accurate
- Sol: Oversampling minority class (SMOTE), undersampling, class weights

Multi-collinearity:

Occurs when two or more independent variables are highly correlated with each other.

- Example: 'Years of Experience' and 'Age' are both related to salary — including both is redundant

Why it matters in Linear Regression:

- In $y = mx + c$, each x variable should contribute independently to y .
- Correlated variables (x_1, x_2) amplify each other's effect and distort coefficients.
- Makes the model unstable and hard to interpret.
- Sol: Remove one of the correlated variables, use PCA (dimensionality reduction), or apply regularization (Ridge/Lasso).